

CONDENSED MATHEMATICS MEETS TOPOLOGICAL QUANTUM FIELD THEORY: A FORMALLY VERIFIED FOUNDATION

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ABSTRACT. We present a formally verified investigation into using condensed abelian groups (Clausen–Scholze) as a target category for topological quantum field theories, replacing topological vector spaces which fail to be abelian. Over six sessions of AI-assisted formal verification using Lean 4, Mathlib v4.28.0 [The24], and Harmonic’s Aristotle system [ABD⁺25], we establish: (1) a sorry-free symmetric monoidal structure on CondensedAb via sheaf localization; (2) a sorry-free abstract TQFT transfer theorem; (3) a sorry-free embedding functor $\text{SemiNormedGrp} \rightarrow \text{CondensedAb}$ (the first bridge between Mathlib’s normed space and condensed libraries); (4) a sorry-free proof of faithfulness for this embedding, together with an elementary argument (not formalized) that it fails to be full; and (5) sorry-free commutative Frobenius algebras and a combinatorial 2d cobordism category.

The project comprises approximately 1160 active lines of Lean 4 across 4 files, with 5 remaining sorries (all with documented proof strategies) and 0 custom axioms. The accompanying formalization is available as a self-contained Lean 4 project.

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Formal verification assisted by Aristotle (Harmonic). Mathematical synthesis and prompt engineering by Claude (Anthropic).

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1. INTRODUCTION

1.1. The Categorical Obstruction in Infinite-Dimensional TQFT. A topological quantum field theory, in Atiyah’s formulation [Ati88], is a symmetric monoidal functor $Z : \text{Cob}_{d+1} \rightarrow \mathcal{C}$ from a cobordism category to some symmetric monoidal target category $\mathcal{C}$. For finite-dimensional state spaces (Dijkgraaf–Witten, Reshetikhin–Turaev [RT91], and Turaev–Viro theories), the target $\mathcal{C} = \text{Vect}$ works perfectly.

The physically interesting theories force infinite-dimensional state spaces. Chern–Simons theory [Wit88] with non-compact gauge group $\text{SL}(2, \mathbb{C})$, quantum gravity partition functions, and path integrals over infinite-dimensional moduli all produce state spaces $Z(\Sigma)$ that cannot live in finite-dimensional Vect . The natural target becomes topological vector spaces.

Here a precise categorical obstruction arises. The category TopAb of topological abelian groups is *not abelian*: a continuous bijective homomorphism need not have a continuous inverse, so a morphism can be simultaneously monic and epic without being an isomorphism. This propagates through the homological algebra machinery. Exact sequences break, derived functors misbehave, and the TQFT gluing axiom (which decomposes state spaces via exact sequences when cutting manifolds) becomes ill-defined.

The completed tensor product compounds the problem. It is not exact: tensoring a short exact sequence with a topological vector space can destroy exactness. Since the TQFT tensor product (disjoint union of boundaries corresponds to tensor product of state spaces) requires this operation, infinite-dimensional TQFTs built on TopVect are categorically deficient.

1.2. The Condensed Fix. Clausen and Scholze’s condensed mathematics [Sch19] resolves this obstruction. The category of condensed abelian groups (sheaves of abelian groups on the coherent site of compact Hausdorff spaces) is an abelian category with all limits and colimits. Every morphism has a kernel and cokernel. Every monic epic is an isomorphism.

Liquid vector spaces, a refinement of condensed modules, admit an exact completed tensor product. This was the content of the Liquid Tensor Experiment [Sch22, CT⁺22], Lean-verified by Commelin et al. in 2022. The exactness of this tensor product is precisely the property TopVect lacks and TQFT gluing requires.

1.3. This Paper. We report on a formal verification effort investigating whether CondensedAb can serve as a TQFT target category. The work was conducted iteratively over six sessions using Harmonic’s Aristotle system [ABD⁺25] for Lean 4 proof search, with Claude (Anthropic) providing mathematical synthesis, prompt engineering, and analysis.

The project produced both expected confirmations and genuine mathematical surprises. The monoidal structure on CondensedAb turned out to already exist in Mathlib (we just had to find it). The embedding from seminormed groups turned out not to be full: an elementary counterexample, found while attempting to formalize fullness, shows the obstruction. (This counterexample is an

ordinary mathematical argument; we did not formalize it in Lean.) Both findings reshaped the project’s direction in real time.

1.4. Navigating the Mathematical Landscape. To contextualize where this work sits within modern mathematics, we developed MathFlowChart, an interactive dependency map of 221 mathematical concepts and 491 connections (available as supplementary material). The chart visualizes the path from foundational structures through the Scholze program to TQFT, tracking the Langlands program connections and chromatic homotopy theory bridges.

2. THE FORMALIZATION: SIX SESSIONS

2.1. Session 1: Abstract TQFT Framework. The first session established the categorical scaffolding in the file `LiquidTQFT.lean`.

Definition 2.1. An *abstract TQFT* with source category S and target category C (both braided monoidal) is a braided monoidal functor $Z : S \rightarrow C$.

```
structure AbstractTQFT
  (S : Type*) [Category S] [MonoidalCategory S] [BraidedCategory S]
  (C : Type*) [Category C] [MonoidalCategory C] [BraidedCategory C]
  where
  Z : S → C
  monoidal : Z.Monoidal
  braided : Z.Braided
```

We work at the level of braided (not symmetric) monoidal functors. This is strictly more general than Atiyah’s original symmetric formulation and covers non-symmetric examples relevant to 3-dimensional TQFTs (e.g., Reshetikhin–Turaev invariants [RT91] from modular tensor categories). Lurie’s cobordism hypothesis [Lur09] provides the fully extended framework in which these structures are most naturally understood.

Theorem 2.2 (Transfer, sorry-free). *If T is a TQFT valued in C and $F : C \rightarrow D$ is a braided monoidal functor, then $T \circ F$ is a TQFT valued in D .*

```
def AbstractTQFT.transfer
  (T : AbstractTQFT S C)
  (F : C → D) [hF : F.Monoidal] [hFb : F.Braided] :
  AbstractTQFT S D where
  Z := T.Z ∘ F
  monoidal := by letI := T.monoidal; infer_instance
  braided := by letI := T.monoidal; letI := T.braided; infer_instance
```

The proof is Lean’s typeclass inference composing the monoidal and braided structures. Additional sorry-free results from this session: `CondensedAb` is abelian, has kernels, cokernels, and finite (co)limits, and satisfies the balanced category property ($\text{mono} + \text{epi} = \text{iso}$).

The session also recorded structural observations about gluing: in any abelian category, short exact sequences provide the decomposition structure needed for TQFT gluing. These take exactness as a hypothesis. The substantive analytic claim (that the liquid tensor product is exact) depends on the LTE and is not verified in this codebase.

2.2. Session 2: Manual Monoidal Construction. The second session (`CondensedMonoidal.lean`) attempted to build `MonoidalCategory CondensedAb` directly, defining the tensor product as sheafification of the pointwise tensor:

$$F \otimes_{\text{cond}} G = \text{sheafify}(F \otimes_{\text{presheaf}} G).$$

This yielded a complete `MonoidalCategoryStruct` (sorry-free) and 5/10 `MonoidalCategory` axioms. The remaining 5 reduced to two “sheafification comparison” isomorphisms:

$$\text{sheafify}(P \otimes Q) \cong \text{sheafify}(\text{sheafify}(P)^{\text{val}} \otimes Q).$$

This analysis was essential for the breakthrough in Session 3. As this manual construction was entirely superseded, it is not included in the final repository; we describe it here because the failed attempt is what motivated the search that found the localization approach.

2.3. Session 3: The Localization Breakthrough. The third session discovered that `Mathlib.CategoryTheory.` already contained the complete infrastructure.

Theorem 2.3 (Sorry-free). *CondensedAb is a symmetric monoidal category.*

The construction assembles four existing Mathlib components:

- (1) **Presheaf monoidal structure:** pointwise tensor on $\text{CompHaus}^{\text{op}} \rightarrow \text{Ab}$.
- (2) **Sheafification as localization:** `presheafToSheaf` is a localization functor for the class $J.W$ of local isomorphisms.
- (3) **W is monoidal:** $J.W$ is closed under tensoring, proved via the internal hom (monoidal closed structure), not stalks.
- (4) **Localized monoidal category:** general construction from [Thea].

```
instance condensedAb_W_isMonoidal :
  ((coherentTopology CompHaus).W
   (A := ModuleCat (ULift   ))).IsMonoidal :=
  GrothendieckTopology.W.monoidal

instance condensedAb_monoidalCategory :
  MonoidalCategory CondensedAb :=
  Sheaf.monoidalCategory _ _
```

Six lines, zero sorries. The internal hom argument (rather than stalks) powers the `W.IsMonoidal` proof, meaning the result holds for any closed braided target category. A direct stalk-based approach has since been formalized by Riou (Mathlib PR #35584), providing an alternative proof path.

2.4. Session 4: The Banach Embedding. The fourth session (`BanachEmbedding.lean`) audited Mathlib for connections between normed spaces and condensed mathematics. The finding: *zero existing bridge*.

Definition 2.4. For a seminormed abelian group V , the presheaf `banachPresheaf V` sends $S \in \text{CompHaus}$ to $\mathcal{C}(S, V)$, the abelian group of continuous maps, with precomposition as functorial action.

Theorem 2.5 (Sorry-free). *banachPresheaf V is a sheaf for the coherent topology on CompHaus.*

The proof transfers the sheaf condition from the Type-valued `yonedaPresheaf` through the forgetful functor. The main technical obstacle was universe arithmetic, bridged using `hasLimitsOfSizeShrink` and `preservesLimitsOfSize_of_univLE`.

Theorem 2.6 (Sorry-free). *There exists a functor $\text{SemiNormedGrp} \rightarrow \text{CondensedAb}$.*

2.5. Session 5: Properties and a Counterexample.

Theorem 2.7 (Sorry-free). *The embedding $\text{SemiNormedGrp} \rightarrow \text{CondensedAb}$ is faithful.*

The faithfulness proof is machine-checked: constant functions at the one-point space separate the points of any seminormed group, so distinct bounded maps induce distinct condensed morphisms.

Proposition 2.8 (Not formalized). *The embedding $\text{SemiNormedGrp} \rightarrow \text{CondensedAb}$ is **not** full.*

Proof. Let $V = c_0(\mathbb{N}, \mathbb{Z})$ (eventually-zero integer sequences with sup norm) and $W = \mathbb{Z}$. The summation map $f(a) = \sum a_n$ is additive and continuous (since $\|a\| < 1$ implies $a = 0$ for integer sequences), but not bounded ($\|e_1 + \dots + e_n\| = 1$ while $f(e_1 + \dots + e_n) = n$). Postcomposition by f defines a valid condensed morphism with no lift to `SemiNormedGrp`. $\square$

We state this as an ordinary mathematical proposition rather than a Lean theorem: unlike the faithfulness result, the failure of fullness is recorded as a paper argument and is not part of the machine-checked development.

Root cause: continuous additive maps between seminormed abelian groups need not be bounded. The standard proof that “continuous linear implies bounded” requires scalar multiplication by a dense subfield. Fullness holds when restricted to `NormedSpace R` for R a nontrivially normed field.

The session also constructed `SemiNormedGrp.hasFiniteProducts` (sorry-free, not previously in Mathlib).

2.6. Session 6: Combinatorial Cobordisms. The sixth session (`Cob2.lean`) formalized the algebraic structure classifying 2-dimensional TQFTs [Koc03].

Definition 2.9 (Sorry-free). A *Frobenius object* in a monoidal category extends Mathlib’s monoid and comonoid objects with the Frobenius relation:

$$(\mu \otimes \text{id}) \circ (\text{id} \otimes \delta) = \delta \circ \mu.$$

A *commutative Frobenius object* additionally requires $\mu \circ \beta = \mu$.

Theorem 2.10 (Sorry-free). *In any braided monoidal category, the unit object carries a canonical commutative Frobenius algebra structure.*

The 2-dimensional cobordism category `Cob2` is defined as a category on the natural numbers, with morphisms generated by μ (pants), η (cap), δ (copants), ε (cup), plus composition, tensor, and swap, quotiented by the Frobenius relations.

3. RESULTS SUMMARY

Result	File	Status
CondensedAb is abelian	LiquidTQFT.lean	Sorry-free
Mono + epi = iso	LiquidTQFT.lean	Sorry-free
MonoidalCategory CondensedAb	LiquidTQFT.lean	Sorry-free
BraidedCategory CondensedAb	LiquidTQFT.lean	Sorry-free
SymmetricCategory CondensedAb	LiquidTQFT.lean	Sorry-free
AbstractTQFT.transfer	LiquidTQFT.lean	Sorry-free
W.IsMonoidal for coherent topology	MonoidalViaLoc.lean	Sorry-free
presheafToSheaf is monoidal	MonoidalViaLoc.lean	Sorry-free
banachPresheaf V is a sheaf	BanachEmbedding.lean	Sorry-free
SemiNormedGrp $\rightarrow$ CondensedAb functor	BanachEmbedding.lean	Sorry-free
Embedding is faithful	BanachEmbedding.lean	Sorry-free
Embedding is full	BanachEmbedding.lean	Disproved (on paper)
SemiNormedGrp.hasFiniteProducts	BanachEmbedding.lean	Sorry-free
FrobeniusObj, CommFrobeniusObj	Cob2.lean	Sorry-free
Trivial Frobenius on $\mathbf{1}_C$	Cob2.lean	Sorry-free
Cob_2 category	Cob2.lean	Sorry-free

TABLE 1. Theorem inventory for the machine-checked development. The failure of fullness (Proposition 2.8) is a paper argument, not a Lean result, and is listed separately.

3.1. Theorem Inventory.

3.2. **Remaining Sorries.** Five sorry placeholders remain across three incomplete results:

- (1) `PreservesEqualizers` for the embedding (hard; 4–5 layers of limit transfer).
- (2) `PreservesFiniteProducts` for the embedding (hard; `ContinuousMap.piEquiv` path).
- (3) `CommFrobeniusData.toCob2Functor` (tedious; three placeholders for the functor’s action on morphisms and its two functoriality proofs, all blocked on case analysis over the quotient).

3.3. **Key Findings. Finding 1.** The monoidal structure on `CondensedAb` was already in Mathlib. The iterative process (axiom $\rightarrow$ manual construction $\rightarrow$ localization discovery) demonstrates the difficulty of navigating large formal libraries.

Finding 2. No prior bridge existed between Mathlib’s `Analysis.NormedSpace` and `Condensed` components. The embedding functor is the first such connection.

Finding 3. Fullness fails for `SemiNormedGrp` (Proposition 2.8, an elementary argument we did not formalize). The condensed category sees strictly more morphisms than the normed category.

Finding 4. `SemiNormedGrp.hasFiniteProducts` was not previously in Mathlib.

Finding 5. The classification “2d TQFT = commutative Frobenius algebra” [Koc03] formalizes cleanly from existing `Mon_`/`Comon_` infrastructure.

4. THE MATHFLOWCHART

To contextualize the mathematical dependencies, we developed MathFlowChart: an interactive HTML/SVG/React diagram mapping 221 nodes and 491 edges across modern mathematics, from categorical foundations through the Scholze program to TQFT. The chart is available as supplementary material.

5. ROADMAP FOR FURTHER DEVELOPMENT

5.1. **Near-term.** Preserve finite limits for the embedding; complete the `Cob2` functor; integrate the project files.

5.2. **Medium-term.** Define `NormedSpaceCat` R and prove fullness (which holds for normed spaces over valued fields); construct a concrete Frobenius object in `CondensedAb`; prove right exactness via the open mapping theorem.

5.3. **Long-term.** Formalize the projective tensor product of normed spaces (not in Mathlib); port the LTE exactness result; formalize smooth cobordism categories.

5.4. **Frontier.** Concrete Chern–Simons constructions valued in `CondensedAb`; Costello–Gwilliam [CG17] factorization algebras in the condensed setting.

6. METHODOLOGY

The work was conducted in six sessions of AI-assisted formal verification. Each session consisted of: (1) a carefully scoped prompt from Claude to Aristotle, based on analysis of previous results; (2) Aristotle producing a Lean 4 file with Mathlib API audit, constructions, and proofs; (3) Claude analyzing the output and designing the next prompt.

The prompts evolved from exploratory to targeted. The most productive prompts front-loaded reconnaissance (searching Mathlib for existing infrastructure) before attempting construction. The Session 3 breakthrough directly resulted from this “search before build” approach.

6.1. Tools.

- **Lean 4** with **Mathlib v4.28.0** [The24] for formal verification.
- **Harmonic Aristotle** [ABD⁺25] for proof search and Mathlib API discovery.
- **Claude (Anthropic)** for mathematical synthesis, prompt engineering, and documentation.
- All formal results independently verifiable via `lake build`.

7. CONCLUSION

Six rounds of AI-assisted formal verification, starting from a blank slate, produced approximately 1160 active lines of Lean 4 across 4 files with 5 sorries and 0 custom axioms. The central results: CondensedAb is a sorry-free symmetric monoidal abelian category, the TQFT transfer theorem is sorry-free, the embedding from seminormed groups exists sorry-free with the sheaf condition verified, and the commutative Frobenius algebra structure classifying 2d TQFTs [Koc03] is sorry-free.

A notable byproduct was identifying that the embedding $\text{SemiNormedGrp} \rightarrow \text{CondensedAb}$, while faithful (proved in Lean), is not full. This is a precise mathematical boundary: condensed abelian groups see more morphisms between normed groups than the normed category itself does, because continuous additive maps need not be bounded without scalar structure. We record this as an elementary argument rather than a formalized result; fullness is recovered by restricting to normed vector spaces over a valued field.

The remaining barriers are increasingly specific: projective tensor products in Mathlib (for the monoidal embedding), porting the LTE exactness result (for full gluing), and smooth cobordism formalization (for geometric TQFTs). The most promising near-term mathematical application is reformulating Costello–Gwilliam [CG17] factorization algebras in the condensed setting.

Code Availability. The complete Lean 4 formalization is available at <https://github.com/seanm27101/Liquid-tft-lean-CM4CM-> and builds against Mathlib v4.28.0. The MathFlowChart interactive dependency map is included as supplementary material.

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